

MLBD MRes PROJECTS 2025-26

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The Project Report and Layperson's Summary – Guidance Note

Deadlines

- **Report with Layperson's Summary:** September 4th at 23:59 pm, electronically in Blackboard
- **Viva:** To be completed before the September 18th **only after handing in your report.**

We suggest submitting in plenty of time before the deadlines to avoid last minute panic. You can submit multiple times before the deadline, it is only the last version that counts. In the unlikely case that you have an approved extension, and you wish to resubmit after the deadline, get in contact with the coordinator. You must ensure that your project report (including layperson's summary) is correctly uploaded to blackboard via the TurnItIn link – **it is your responsibility to ensure the submission is successful.**

You may find the ECRI writing sessions useful while writing your final report:

<https://www.imperial.ac.uk/early-career-researcher-institute/learning-and-development/writing-sessions/>

Introduction

The MRes project offers an opportunity to develop research skills and one of the most important skills is the ability to communicate your work clearly in a written report. The MRes projects cover a broad range of computational and experimental topics which differ greatly in scientific scope and potential. Hence, a few projects may lead to new results which can be published, and some may fail to work out, and the rest lie somewhere in-between. This is not unsurprising as research is about exploring the unknown. Therefore, the assessment of the report will not be based on whether or not some genuinely new result was obtained; it will rather be assessed on the efforts you made to tackle your research topic, the appropriateness of the methods that you explored and applied, and the clarity and completeness of your explanation of this topic and what you have learned from it.

You should keep in mind that your report and layperson's summary will be independently assessed by your project supervisor and project assessor. It is very important that **your report should be aimed at a physicist who is not an expert in the area of your project.** Any reasonably experienced physicist should be able to read through your Project Report and understand it. **Be careful to introduce concepts and terms that were new to yourself when you started the project and avoid jargon that is used within the specific area where you worked.**

Report outline

The report outline can be maximally three pages long. It should be used to receive feedback on the structure and style of the final report. It is suggested to include an outline structure of the individual sections, maybe some sample figures or a section where you try to write everything. You should discuss the project outline with your supervisor and your supervisor should provide feedback on this before you begin writing your report in full. The outline is not assessed.

Full Report

Word and page limits

The MRes Project Report should be typeset in single column format. If you use Word, LaTeX or something else for this doesn't matter, but you should be able to produce a PDF file in the end. **The font size should be 12pt.** The margins should not be smaller than 2 cm (as used in this document). The page size should be A4.

The MRes Project Report should be around 8,000 - 10,000 words long. The word count excludes title page, layperson's summary, contents page (if included), acknowledgements, and bibliography. **Everything else is included, i.e. abstract, main body, figure captions, table captions and appendices.** The content of tables themselves, mathematical formulae and small extracts of computer code is not included.

Typically, this is about 25-30 A4 pages in length including figures and tables but excluding title page, contents page(s) and bibliography.

The report may not exceed 10,000 words. If your supervisor recommends otherwise, please get in contact with coordinator!

Structure of the Report

The report should be concise and complete. In structure, it should have:

- **Title page.**

A title page with:

1. **A title that clearly identifies what the report is about. The title is not required to be the same as the title of the original project proposal.**
2. **The project code (e.g. MLBD_01)**
3. **Your name**
4. **Your CID Number**
5. **The name of the supervisor**
6. **The name of the assessor**
7. **The word count (not including the layperson's summary)**
8. **The date of submission**
9. **The Imperial logo, but not the College Crest (to comply with college policy).**

You must use one of the Word (.docx) or Latex (.tex) front page templates for your title page, that have been provided in Blackboard.

- **Layperson's Summary**

The layperson's summary is a one-page article that describes the MRes project and its significance in a way that is engaging and accessible to a typical A-level student. See specific guidance for the layperson's summary at the end of this document.

- **Abstract**

The abstract should clearly and concisely identify the principal features of the work and the results achieved and the conclusions that can be drawn, if any.

- **Contents**

The contents pages list the chapters, sections, and subsections. You can include this if you find it useful, but it is not required.

- **Introduction**

The MRes Project Report should have an introductory chapter in which the aims and objectives of the project are established and in which the work is clearly put into context. The introductory chapter will be very important for the assessor. It should contain an overview of the project and should be readily understood by any physicist.

- **Discussions of the methods/equipment/ML theory employed**

The methods may be experimental and/or computational as appropriate to the MRes Project.

- **Results**

A clear description and presentation of the results/observations obtained. Think carefully how to convey the results with clarity and in the most informative way. Figures must have clearly labelled axes and scales and an extensive figure caption that states what is plotted and stresses what the reader should notice in particular. Make sure that the text on figures is not too small.

- **Discussion of results**

A critical discussion of the results and the experimental or computational errors (eg accuracy of predictive models and uncertainties associated with evaluating this accuracy) and the conclusions which can be drawn from the results. For the main part of the report (so this and the previous two parts), there can be variation in the structure that makes it fit better with the actual project. Use your common sense.

- **Conclusions**

In your conclusions you should discuss to what extent the aims of the project which you gave earlier have been achieved and what further work would be appropriate.

- **Acknowledgements**

In a formal acknowledgement section refer to the people you wish to thank.

- **Bibliography**

You must provide a clear and extensive set of references (bibliography) to which you have referred in the text. These should be in the standard format used in scientific journals. Take care to avoid incomplete references where the journal name, edition, year, authors or other parts are missing. It is important to be consistent in style for both how you cite within the main text and in the formatting of the bibliography items.

- **Appendices (if appropriate).**

Material that somehow breaks the flow if included in the main report. Can be lengthy mathematical derivations, computational algorithms, of reference figures. **The appendices are included in the word count. Do not** include pages and pages of code in the appendices (or in the main body). Instead, you should include references to code in online repositories such as the Imperial College GitHub Enterprise, in your bibliography.

It is most important that a reasonably experienced scientist should be able to read through your MRes Project Report and immediately be able to follow what you did, to put your work into context, and from your report be able to carry out work similar to your MRes Project. Hence, it must contain detailed descriptions of any equipment used, datasets used, links or references to software packages and generally anything which would help you in repeating the work again in 5 years. Note that while there is no official requirement for all code to be made available, it

is a good idea to place it on GitHub (or similar) and then refer to it from within the report so that the assessors can verify additional work done for the project.

Reports must be complete and concise and therefore avoid waffle. Also make sure that you back up any points you make. We leave it to you to make sure your MRes Project Report is of a high quality and “professionally” written. This means you also should be careful about proofreading to avoid spelling errors, typos and poor grammar. **Make sure to consider the feedback you were given on your literature review and report outline in this aspect.**

It is important for you to realise that most projects are never finished and so at the time you write your report, you will only be able to cover what has been done by that time. In some cases, it may seem that little has been achieved, however, this is no reason for not producing an excellent report covering many of the points made above.

Plagiarism

Plagiarism must be strictly avoided (penalties for this can be very severe). Therefore, never use the words, software or ideas of others (from whatever source they might arise e.g. published papers, the web, other student reports or content produced from generative AI) without a proper attribution in the form of citations. Remember that even with attribution, you have to write the report in your own words. Note that you cannot directly copy sections of your own Literature Review, without explicit citation of it. Doing so counts as plagiarism under the college’s regulations. Keep in mind that the aim and objectives of your project might have moved since then, such that parts of your literature review might not be relevant.

As a reminder, please see the departmental guidance on the use of generative AI and plagiarism:

<https://www.imperial.ac.uk/media/imperial-college/faculty-of-natural-sciences/departments-of-physics/public/students/current-students/pgt/Generative-AI-Tools-v1.1.pdf>

For further information on plagiarism, see

www.imperial.ac.uk/admin-services/library/learning-support/plagiarism-awareness/

You are required to submit the report electronically (TurnItIn) to be run through software to check for plagiarism.

Back-ups

Keep electronic back-ups of your report to avoid losing your work.

Marking criteria

The report is marked using the criteria below with their respective relative weights.

Quality of the report

- Structure and language (20%)
- Clarity (20%)

Quality of the science

- Techniques (25%)
- Scientific achievements (25%)

Layperson’s Summary (10%)

Below, you see the guidance on the individual categories that is given to your supervisor and assessor for marking the report. **Keep these in mind when writing your report.**

Structure and language

How well does the report read as a scientific report? For this mark, you should look at

- Spelling and grammar
- Consistency of notation
- Typographical quality of figures, equations and tables (how they look)
- Typesetting of bibliography (not if they are relevant) and consistent citation method
- Use of sections
- Logic of structure

Clarity

The report is intended for a general audience and not for experts in the field. The guidance to the students says “It is very important that your report should be aimed at a physicist who has not worked in the area of your project. Any reasonably experienced physicist should be able to read through your MRes Project Report and understand it.” For this mark, you should look at

- Ability of non-expert to understand all parts of report, including the definition of all specialist terms used.
- Use of references (not the typesetting of bibliography)
- Choice of figures, equations and tables (what they show, not if they look good)
- Explanation of results.

Techniques

From reading the report, make a judgement of the skills demonstrated. This might include skills in computing, experimental work, data analysis techniques, critical/statistical analysis of published data etc. – whatever is appropriate for the individual project. Consider in this part also the treatment of uncertainties, any possible biases in the results obtained, and an appropriate critique of any particular ML model you have used in your project.

Scientific achievements

Based on the content of the report, evaluate the science that has been carried out. What has the student achieved in the allotted time? Consider both the volume and quality of the work completed. A research project that is not completed may still achieve high marks if the work carried out is excellent within the time constraint.

Layperson’s summary

The Layperson’s summary is a one-page article that describes the MRes project and its significance in a way that is engaging and accessible to a typical A-level student. You can find more guidance on this at the end of this document.

Assessment will focus on these two main points:

1. Does the summary describe the specific research project, including motivation and results?
2. Is the summary pitched at the right level and written in a way that is engaging for the target audience?

Handing in

The report should be submitted online in BlackBoard as a PDF file. Please make sure that you have access by checking a few days before the deadline. **You must name your PDF file as “PROJECTCODE_YOURCID.pdf”** – eg if you are taking the project “MLBD_01” and your CID is “01234567” your pdf must be named “MLBD_01_01234567.pdf”.

Final comment

Keep a copy of your own MRes Project Report. It may be useful in, for instance, job or research studentship applications, even a few years later.

Viva

The viva is done with the supervisor and the assessor, after you have handed in your report. It consists of three parts.

- An oral presentation from you. Typically, this takes shape as a presentation where you have a set of slides. You are given 15 minutes to make your presentation and timeliness is part of the assessment criteria.
- After the oral presentation, the supervisor and the assessor will ask you questions. These can range from very detailed questions on how you implemented something, or they can be on the broader context of your work and its implications.
- Feedback on the viva itself. How you performed, how you could be better in a similar situation in the future and recommendations on what to take particular care with for your final report.

The assessment of the viva is done with equal weight to the following criteria:

- How the work fitted into the context of related work.
- If it presented a comprehensive review of the work undertaken.
- How clear were the conclusions which could be drawn.
- Confidence in delivery of presentation, when discussing the work and answers to questions.
- Presentation quality, that is, the logical structure, the quality of slides, figures and graphs and meeting the 15 min requirement.

You can include additional work in the viva which did not arrive in time for the report.

Guidance on Layperson's summary

Summarise your MRes project in a one-page article suitable for a A-level student to understand and enjoy

Brief Overview

Your goal in this activity is to write a one-page article describing your MRes project in a way that is accessible to a reader who is not an expert in the field but has some scientific understanding (an A-level student). The article should be engaging, written in your own personal style, and should seek to inspire your audience by making it clear what the impact of your work could ultimately be.

Take a look at articles from Magazines like "New Scientist" or articles from reputable news outlets that often have Science and technology sections. These will help you find the right style for your audience. You can combine this with your own personal experience of reading popular science journals or books (and any similar writing you may have already done) to pitch your article correctly. You can also take a look at our own Imperial "outreach" articles: (<https://www.imperial.ac.uk/physics/engage-with-us/outreach/>) for an idea about the language / style to use. Remember to make it engaging and exciting!

Specific Requirements

The article must be no longer than one page of A4, word count less than 700 words.

You will submit the summary as part of your MRes Report.

While you will need to describe some general aspects of your area of research, the article must focus on your specific project, including motivation, methods, results and impact.

Make sure you include a title! This can be different than your MRes Report title.

The inclusion of figures is likely to be beneficial but if they are included, they should be referred to in the text and the source given in the caption (if it's not your own figure). Make sure text in the figure is legible. References in the text are discouraged but you can name institutions or specific people in the text, if appropriate.

General Comments

This activity will give you experience in writing for a non-specialist but scientifically literate audience. You will need to find ways to explain technical terms and concepts so that your audience is able to understand your work and its potential significance. You must avoid using jargon.

This skill is something that most physicists find difficult yet is hugely beneficial for both effective outreach and, as you have seen, can help to secure funding. More generally, being able to tailor your written (and verbal) presentation for a specific target audience ensures effective communication in the workplace and elsewhere.

Skills developed/tested

Ability to concisely summarise research and explain key findings and their significance in an engaging way to a lay audience through writing

Marking

Your mark will focus on these two main points:

1. Does the summary describe the specific research project, including motivation and results?
2. Is the summary pitched at the right level and written in a way that is engaging for the target audience?